Class 18

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CMI-PB Project **6**

Focus on "PT" Pertussis Toxin antigen 16

Background

Pertussis (a.k.a Whooping Cough) is a common lung infection caused by the bacteria B. Pertussis.

This can infect people of any age but is most deadly in infants due to their small airways and weakened immune response.

The CDC tracks the number of reported cases in the US We can "scrape' this data with the **datapasta** package.

```
cdc <- data.frame(
                                 Year = c(1922L,1923L,1924L,1925L,
                                          1926L,1927L,1928L,1929L,1930L,1931L,
                                          1932L,1933L,1934L,1935L,1936L,
                                          1937L,1938L,1939L,1940L,1941L,1942L,
                                          1943L,1944L,1945L,1946L,1947L,
                                          1948L,1949L,1950L,1951L,1952L,
                                          1953L,1954L,1955L,1956L,1957L,1958L,
                                          1959L,1960L,1961L,1962L,1963L,
                                          1964L,1965L,1966L,1967L,1968L,1969L,
                                          1970L,1971L,1972L,1973L,1974L,
                                          1975L,1976L,1977L,1978L,1979L,1980L,
                                          1981L,1982L,1983L,1984L,1985L,
                                          1986L,1987L,1988L,1989L,1990L,
```

```r
                                        1991L,1992L,1993L,1994L,1995L,1996L,
                                        1997L,1998L,1999L,2000L,2001L,
                                        2002L,2003L,2004L,2005L,2006L,2007L,
                                        2008L,2009L,2010L,2011L,2012L,
                                        2013L,2014L,2015L,2016L,2017L,2018L,
                                        2019L,2020L,2021L,2022L,2023L,2024L,2025L),
                                Cases = c(107473,164191,165418,152003,
                                        202210,181411,161799,197371,
                                        166914,172559,215343,179135,265269,
                                        180518,147237,214652,227319,103188,
                                        183866,222202,191383,191890,109873,
                                        133792,109860,156517,74715,69479,
                                        120718,68687,45030,37129,60886,
                                        62786,31732,28295,32148,40005,
                                        14809,11468,17749,17135,13005,6799,
                                        7717,9718,4810,3285,4249,3036,
                                        3287,1759,2402,1738,1010,2177,2063,
                                        1623,1730,1248,1895,2463,2276,
                                        3589,4195,2823,3450,4157,4570,
                                        2719,4083,6586,4617,5137,7796,6564,
                                        7405,7298,7867,7580,9771,11647,
                                        25827,25616,15632,10454,13278,
                                        16858,27550,18719,48277,28639,32971,
                                        20762,17972,18975,15609,18617,
                                        6124,2116,3044,7063,22538,21996)
       )
cdc
```

```
    Year  Cases
1   1922 107473
2   1923 164191
3   1924 165418
4   1925 152003
5   1926 202210
6   1927 181411
7   1928 161799
8   1929 197371
9   1930 166914
10  1931 172559
11  1932 215343
12  1933 179135
13  1934 265269
```

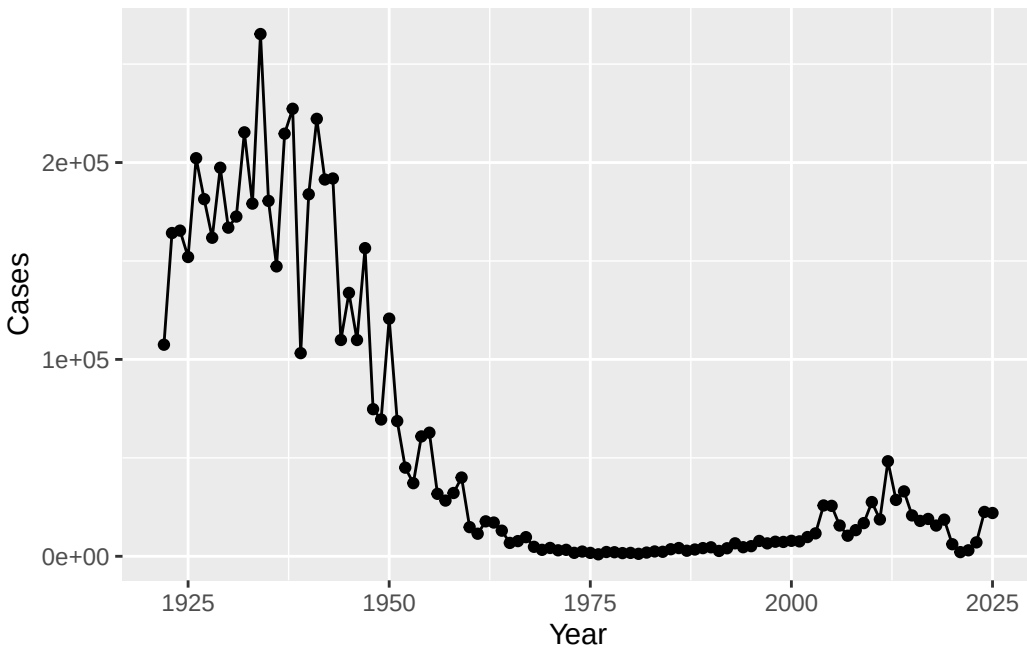
14	1935	180518
15	1936	147237
16	1937	214652
17	1938	227319
18	1939	103188
19	1940	183866
20	1941	222202
21	1942	191383
22	1943	191890
23	1944	109873
24	1945	133792
25	1946	109860
26	1947	156517
27	1948	74715
28	1949	69479
29	1950	120718
30	1951	68687
31	1952	45030
32	1953	37129
33	1954	60886
34	1955	62786
35	1956	31732
36	1957	28295
37	1958	32148
38	1959	40005
39	1960	14809
40	1961	11468
41	1962	17749
42	1963	17135
43	1964	13005
44	1965	6799
45	1966	7717
46	1967	9718
47	1968	4810
48	1969	3285
49	1970	4249
50	1971	3036
51	1972	3287
52	1973	1759
53	1974	2402
54	1975	1738
55	1976	1010
56	1977	2177

57	1978	2063
58	1979	1623
59	1980	1730
60	1981	1248
61	1982	1895
62	1983	2463
63	1984	2276
64	1985	3589
65	1986	4195
66	1987	2823
67	1988	3450
68	1989	4157
69	1990	4570
70	1991	2719
71	1992	4083
72	1993	6586
73	1994	4617
74	1995	5137
75	1996	7796
76	1997	6564
77	1998	7405
78	1999	7298
79	2000	7867
80	2001	7580
81	2002	9771
82	2003	11647
83	2004	25827
84	2005	25616
85	2006	15632
86	2007	10454
87	2008	13278
88	2009	16858
89	2010	27550
90	2011	18719
91	2012	48277
92	2013	28639
93	2014	32971
94	2015	20762
95	2016	17972
96	2017	18975
97	2018	15609
98	2019	18617
99	2020	6124

```
100 2021    2116
101 2022    3044
102 2023    7063
103 2024   22538
104 2025   21996
```

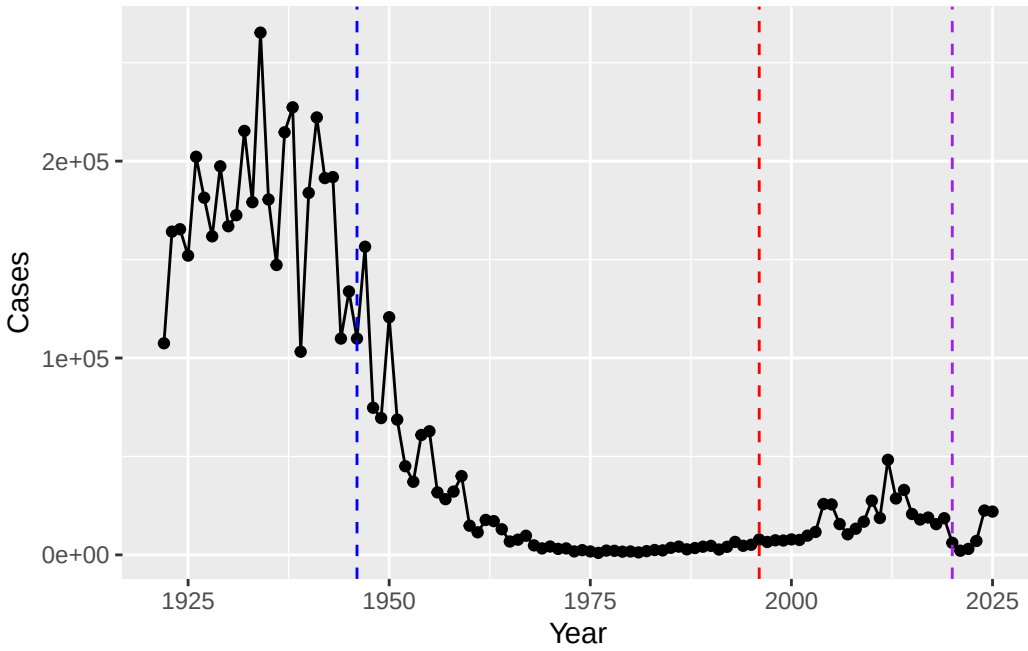
Q. Make a plot of `year` vs `cases`

```
library(ggplot2)
 ggplot(cdc) + aes(Year, Cases) + geom_line() + geom_point()
```



Q. Add some major milestones including he first wP vaccine rollout(1946), the switch to the newer aP vaccine (1996), and the COVID years(2020)

```
ggplot(cdc) + aes(Year, Cases) + geom_line() + geom_point() +
 geom_vline(xintercept = 1946, col="blue", lty =2, ) +
 geom_vline(xintercept = 1996, col="red", lty =2 , ) +
   geom_vline(xintercept = 2020, col="purple", lty =2)
```



In 1945 the wP vaccine was introduced causing cases rates to begin to drop dramatically. In 1996 the alternative aP vaccine was which wasn't as effective as the wP vaccine leading to the rise of cases and the need for booster shots. Additionally, during the early 2000's the anti-vax movement began to gain major traction causing less people to give the vaccine to their children and leading to a rise in cases. In 2020, protocols desgined to limit the spread of COVID also limited the spread of other transmissible diseases.

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB Project

Computational Models of Immunity Pertssis Boostproject aims to provide the scientificcommunity with this very information.

They make their data vailable via JSON format APIs.we can read this in R with the `read_json()` function from the **jsonlite** package.

```
library(jsonlite)
 subject <- read_json("http://cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
```

```r
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many "wP" and "aP" individuals are i this `subject` table.

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the biological sex breakdown?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. In terms of race and gender is this dataset representative of the US population?

```r
table(subject$biological_sex, subject$race, subject$ethnicity)
```

```
, ,  = Hispanic or Latino


        American Indian/Alaska Native Asian Black or African American
  Female                            0     0                        0
  Male                              0     0                        0

        More Than One Race Native Hawaiian or Other Pacific Islander
  Female                  9                                        0
  Male                    2                                        0

        Unknown or Not Reported White
  Female                     11    11
  Male                        4     4

, ,  = Not Hispanic or Latino


        American Indian/Alaska Native Asian Black or African American
  Female                            0    32                        2
  Male                              1    12                        3

        More Than One Race Native Hawaiian or Other Pacific Islander
  Female                  6                                        1
  Male                    2                                        1

        Unknown or Not Reported White
  Female                      2    36
  Male                        1    27

, ,  = Unknown


        American Indian/Alaska Native Asian Black or African American
  Female                            0     0                        0
  Male                              0     0                        0

        More Than One Race Native Hawaiian or Other Pacific Islander
  Female                  0                                        0
  Male                    0                                        0

        Unknown or Not Reported White
  Female                      1     1
```

```
  Male                          2     1
```

Let's read some more database tables:

```
specimen <-  read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE
                    )
ab_titler <-  read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```
head(ab_titler)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to join (merge/link) multiple tables so we have all the data in one place and not spread out everywhere.

We can use the `*_join()` family of functions from **dplyr** to do this.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
meta <-  inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
```

```
2                            1                            1         Blood
3                            3                            3         Blood
4                            7                            7         Blood
5                           11                           14         Blood
6                           32                           30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titler, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
```

```
1                           -3                              0         Blood
2                           -3                              0         Blood
3                           -3                              0         Blood
4                           -3                              0         Blood
5                           -3                              0         Blood
6                           -3                              0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. What Antibody isotypes are measured for these patients?

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4
 6698  7265 11993 12000 12000 12000
```

Q12 . What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

Q. What antigens are reported?

```
table(abdata$antigen)
```

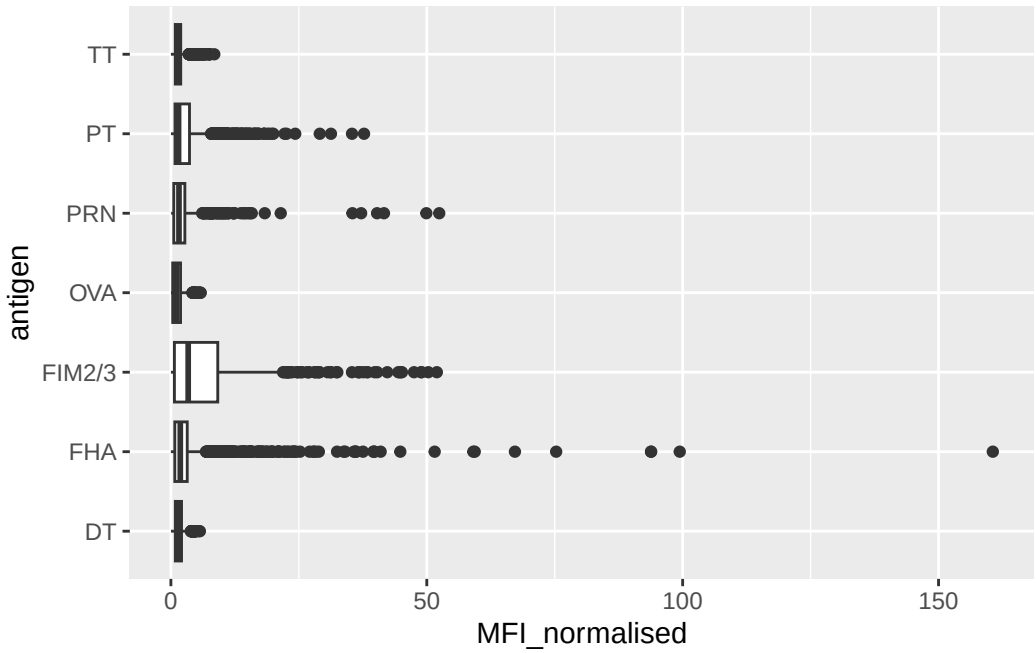
```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
    PD1     PRN      PT     PTM   Total      TT
   1970    6712    6712    1970     788    6318
```

Let's focus on the IgG isotype and make a plot of MFI_normalized for antigens.

```r
IgG <- abdata |>
  filter(isotype=="IgG")
head(IgG)
```

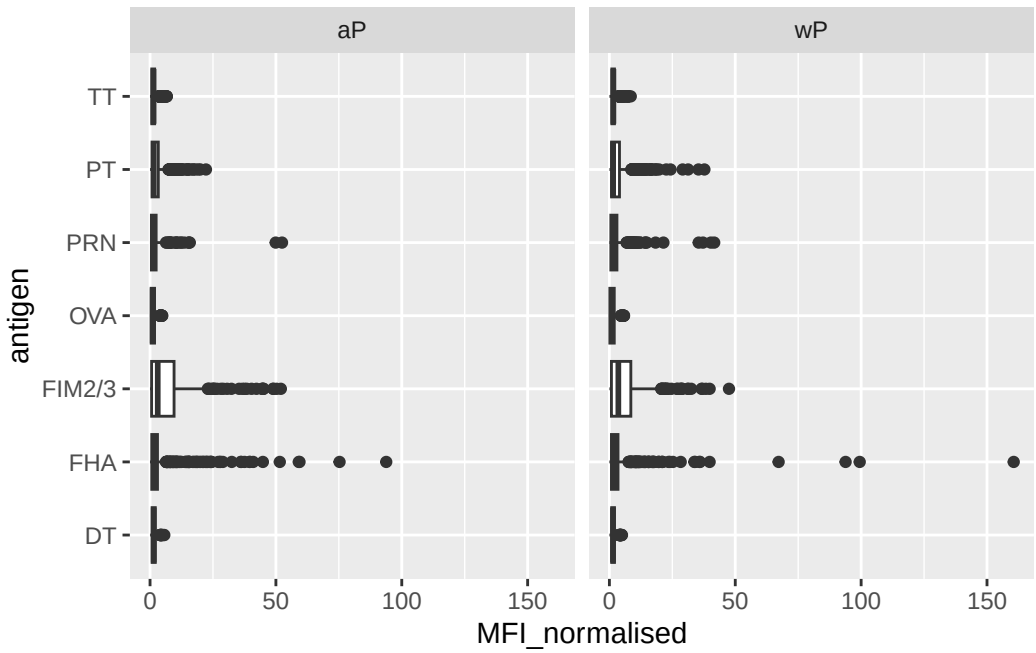
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
              ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4               Unknown White    1983-01-01    2016-10-10 2020_dataset
5               Unknown White    1983-01-01    2016-10-10 2020_dataset
6               Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

```r
ggplot(IgG) +
  aes(MFI_normalised, antigen)+
  geom_boxplot()
```

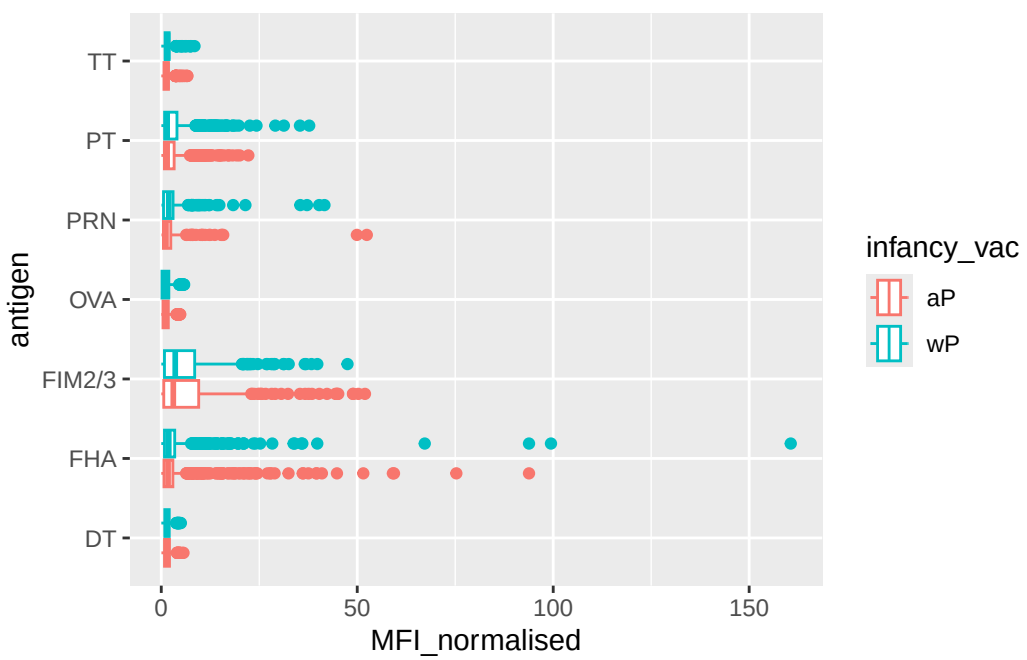


Q. Is there a difference for aP vs. wP individuals with these values?

```r
ggplot(IgG) +
  aes(MFI_normalised, antigen)+
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

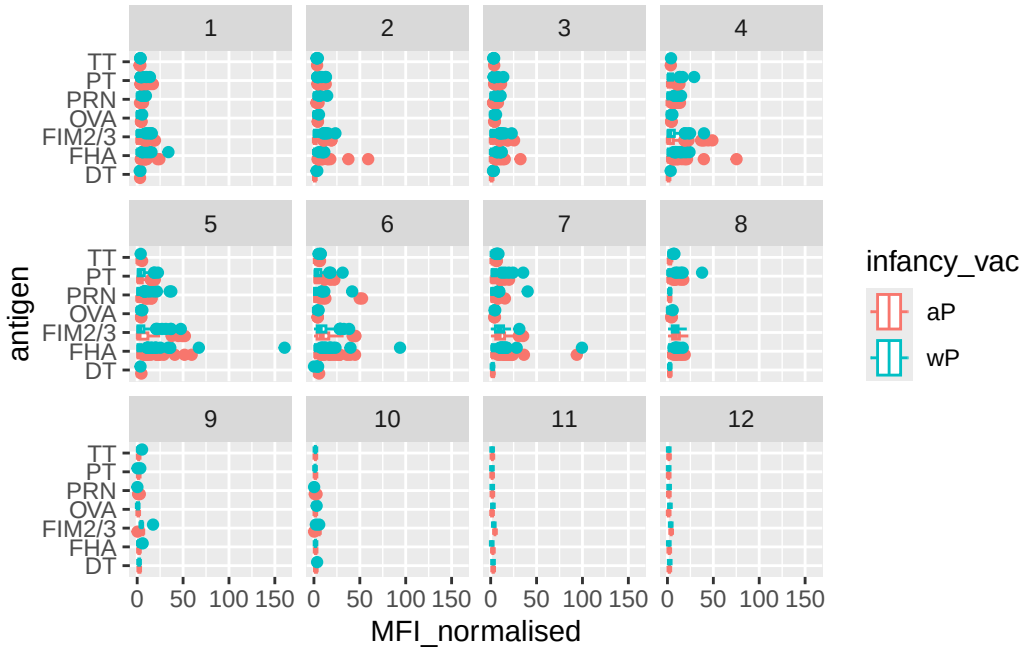



```
ggplot(IgG) +
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```



Q. Is there a temporal response - i.e do values increase or decrease over time?

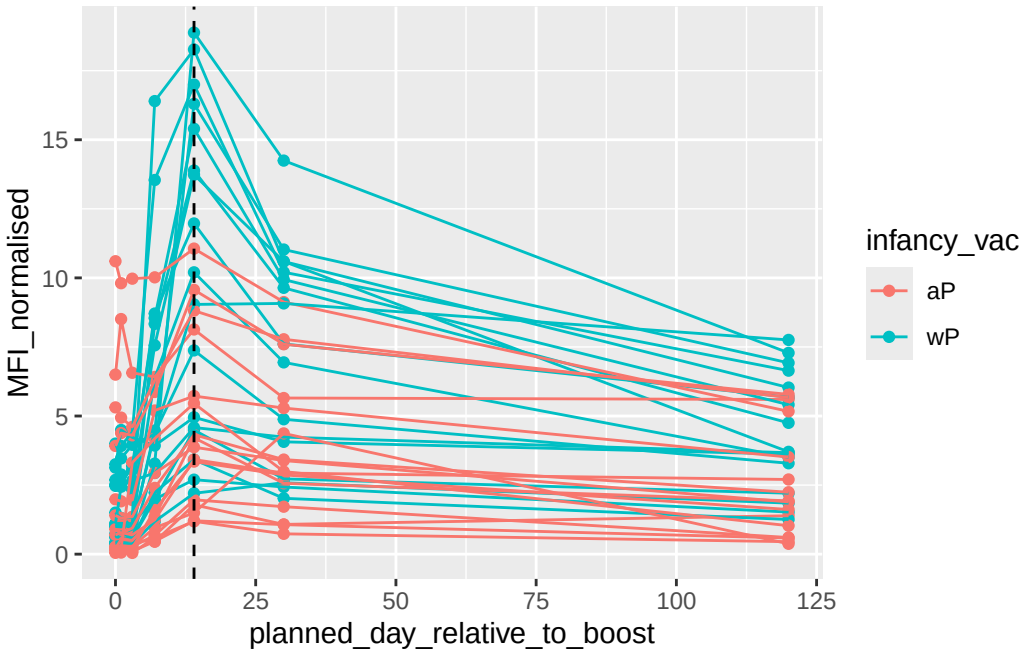
```
ggplot(IgG) +
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot() + facet_wrap(~visit)
```



Focus on "PT" Pertussis Toxin antigen

```
pt.IgG.21 <- IgG |> filter (antigen=="PT",
                dataset== "2021_dataset")
```

```
ggplot(pt.IgG.21) +
  aes(planned_day_relative_to_boost, MFI_normalised, col=infancy_vac, group=subject_id)+
  geom_point() +
  geom_line()+
  geom_vline(xintercept = 14, lty=2)
```

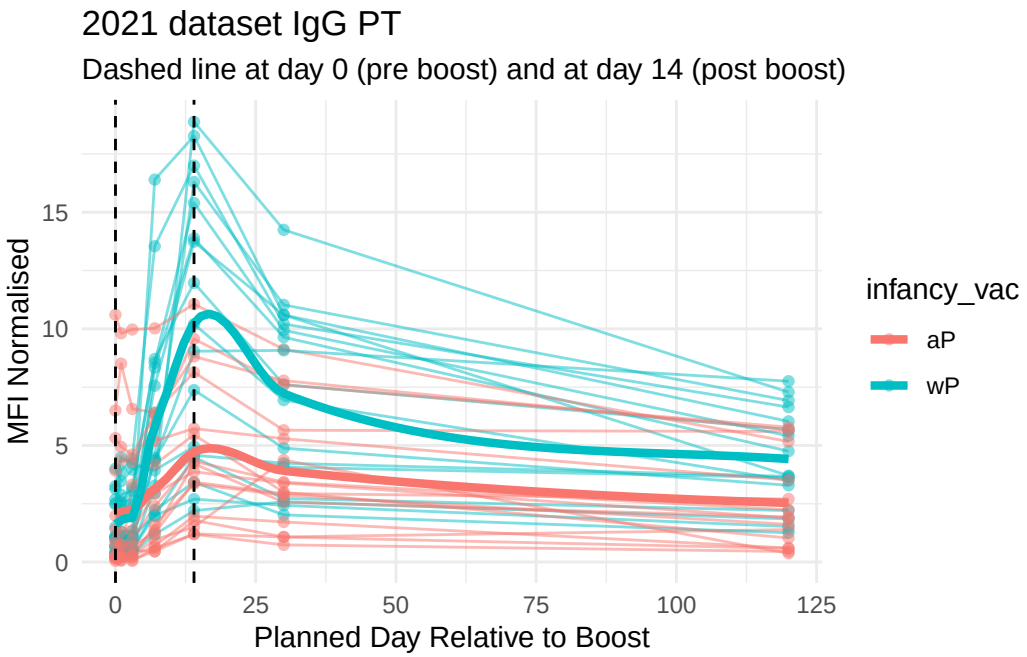



```
pt.IgG.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot(aes(x = planned_day_relative_to_boost,
             y = MFI_normalised,
             col = infancy_vac)) +
  # 1. Individual lines: Keep group=subject_id here, but add transparency (alpha)
  geom_line(aes(group = subject_id), alpha = 0.5) +
  geom_point(aes(group = subject_id), alpha = 0.5) +

  # 2. Average lines: Remove the subject grouping so it averages by infancy_vac
  # Using 'se = FALSE' hides the shaded confidence interval for a cleaner look
  geom_smooth(aes(group = infancy_vac), span=0.4 , method = "loess", se = FALSE, size = 1.5)

  # Your existing vertical lines and labels
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 14, linetype = "dashed") +
  labs(title = "2021 dataset IgG PT",
       subtitle = "Dashed line at day 0 (pre boost) and at day 14 (post boost)",
       x = "Planned Day Relative to Boost",
       y = "MFI Normalised") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```


2021 dataset IgG PT
Dashed line at day 0 (pre boost) and at day 14 (post boost)
MFI Normalised
0
5
10
15
0
25
50
75
100
125
Planned Day Relative to Boost
infancy_vac
aP
wP